

5.1 Algebraic Expressions



Figure 5.2 Two college graduates! (credit: modification of work UC Davis College of Engineering/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Convert between written and symbolic algebraic expressions and equations.
2. Simplify and evaluate algebraic expressions.
3. Add and subtract algebraic expressions.
4. Multiply and divide algebraic expressions.

Algebraic expressions are the building blocks of algebra. While a numerical expression (also known as an arithmetic expression) like $5 + 3$ can represent only a single number, an algebraic expression such as $5x + 3$ can represent many different numbers. This section will introduce you to algebraic expressions, how to create them, simplify them, and perform arithmetic operations on them.

Algebraic Expressions and Equations

Xavier and Yasenia have the same birthday, but they were born in different years. This year Xavier is 20 years old and Yasenia is 23, so Yasenia is three years older than Xavier. When Xavier was 15, Yasenia was 18. When Xavier will be 33, Yasenia will be 36. No matter what Xavier's age is, Yasenia's age will always be 3 years more.

In the language of algebra, we say that Xavier's age and Yasenia's age are variable and the 3 is a constant. The ages change, or vary, so age is a **variable**. The 3 years between them always stays the same or has the same value, so the age difference is the **constant**. In algebra, letters of the alphabet are used to represent variables. The letters most often used for variables are x , y , z , a , b , and c . Suppose we call Xavier's age x . Then we could use $x + 3$ to represent Yasenia's age, as shown in the table below.

Xavier's Age	Yasenia's Age
15	18
20	23
33	36
x	$x + 3$

To write algebraically, we need some symbols as well as numbers and variables. The symbols for the four basic arithmetic operations: addition, subtraction, multiplication, and division are summarized in [Table 5.1](#), along with words we use for the operations and the result.

Operation	Notation	Say:	The result is...
Addition	$a + b$	a plus b	The sum of a and b
Subtraction	$a - b$	a minus b	The difference of a and b
Multiplication	$a \cdot b$, $(a)(b)$, $(a)b$, $a(b)$, ab , ba	a times b	The product of a and b
Division	$a \div b$, a/b	a divided by b	The quotient of a and b

Table 5.1 Symbols for Operations

 In algebra, the cross symbol ($\times$) is normally not used to show multiplication because that symbol could cause confusion. For example, does $3xy$ mean $3 \times y$ (three times y) or $3 \bullet x \bullet y$ (three times x times y)? To make it clear, use $\bullet$ or parentheses for multiplication.

We perform these operations on two numbers. When translating from symbolic form to words, or from words to symbolic form, pay attention to the words *of* or *and* to help you find the numbers.

- The **sum of 5 and 3** means add 5 plus 3, which we write as $5 + 3$.
- The **difference of 9 and 2** means subtract 9 minus 2, which we write as $9 - 2$.
- The **product of 4 and 8** means multiply 4 times 8, which we can write as $4 \bullet 8$.
- The **quotient of 20 and 5** means divide 20 by 5, which we can write as $20 \div 5$.

EXAMPLE 5.1**Translating from Algebra to Words**

Translate the following algebraic expressions from algebra into words.

1. $12 + 14$
2. $(30)(5)$
3. $64 \div 8$
4. $x - y$

 Solution

1. According to Table 5.1, this could be translated as 12 plus 14 OR the sum of 12 and 14.
2. According to Table 5.1, this could be translated as 30 times 5 OR the product of 30 and 5.
3. According to Table 5.1, this could be translated as 64 divided by 8 OR the quotient of 64 and 8.
4. According to Table 5.1, this could be translated as x minus y OR the difference of x and y .

 YOUR TURN 5.1

Translate the following algebraic expressions from algebra into words.

1. $18 + 11$
2. $(27)(9)$
3. $84 \div 7$
4. $p - q$

EXAMPLE 5.2**Translating from Words to Algebra**

Translate the following phrases from words into algebraic expressions.

1. The difference of 47 and 19

2. 72 divided by 9
3. The sum of m and n
4. 13 times 7

 Solution

1. According to [Table 5.1](#), these words could be translated as $47 - 19$.
2. According to [Table 5.1](#), these words could be translated as $72 \div 9$.
3. According to [Table 5.1](#), these words could be translated as $m + n$.
4. According to [Table 5.1](#), these words could be translated as $(13)(7)$.

 YOUR TURN 5.2

Translate the following phrases from words into algebraic expressions.

1. 43 plus 67
2. The product of 45 and 3
3. The quotient of 45 and 3
4. 89 minus 42

What is the difference in English between a phrase and a sentence? A phrase expresses a single thought that is incomplete by itself, but a sentence makes a complete statement. “Running very fast” is a phrase, but “The football player was running very fast” is a sentence. A sentence has a subject and a verb. In algebra, we have **expressions** and equations. [Example 5.1](#) and [Example 5.2](#) used expressions. An expression is like an English phrase. Notice that the English phrases do not form a complete sentence because the phrase does not have a verb. The following table has examples of expressions, which are numbers, variables, or combinations of numbers and variables using operation symbols.

Expression	Words	English Phrase
$3 + 5$	3 plus 5	The sum of three and five
$n - 1$	n minus one	The difference of n and one
$6 \bullet 7$	6 times 7	The product of six and seven
$x \div y$	x divided by y	The quotient of x and y

EXAMPLE 5.3

Translating from an English Phrase to an Expression

Translate the following phrases from words into algebraic expressions.

1. Seven more than a number n .
2. A number n times itself.
3. Six times a number n , plus two more.
4. The cost of postage is a flat rate of 10 cents for every parcel, plus 34 cents per ounce x .

 Solution

1. $n + 7$
2. $n \bullet n$ or n^2
3. $6n + 2$
4. $10 + 34x$

YOUR TURN 5.3

Translate the following phrases from words into algebraic expressions.

1. Twenty less than a number n . (Hint: you have a number n and you want 20 less than it.)
2. Add two to a number n , then multiply it by six.
3. A number n to the third power minus five.
4. A plumber charges \$60 per hour h , plus a \$40 flat fee for every job.

An **equation** is two expressions linked with an **equal sign** (the symbol $=$). When two quantities have the same value, we say they are equal and connect them with an equal sign. When you read the words the symbols represent in an equation, you have a complete sentence in English. The equal sign gives the verb. So, $a = b$ is read " a is equal to b ." The following table has some examples of equations.

Equation	English Sentence
$3 + 5 = 8$	The sum of three and five is equal to eight.
$n - 1 = 14$	n minus one equals fourteen.
$6 \bullet 7 = 42$	The product of six and seven is equal to forty-two.
$x = 53$	x is equal to fifty-three.
$y + 9 = 2y - 3$	y plus nine is equal to two times y minus three.

EXAMPLE 5.4

Translating from an English Sentence to an Equation

Translate the following sentences from words into algebraic equations.

1. Two times x is 6.
2. n plus 2 is equal to n times 3.
3. The quotient of 35 and 7 is 5.
4. Sixty-seven minus x is 56.

Solution

1. $2x = 6$
2. $n + 2 = 3n$
3. $35 \div 7 = 5$
4. $67 - x = 56$

YOUR TURN 5.4

Translate the following sentences from words into algebraic equations.

1. Five times y is 50.
2. Half of a number n is 30.
3. The difference of three times a number n and 7 is 2.
4. Two times x plus 7 is 21.

WHO KNEW?

The Use of Variables

French philosopher and mathematician René Descartes (1596–1650) is usually given credit for the use of the letters x , y , and z to represent unknown quantities in algebra. He introduced these ideas in his publication of *La Geometrie*, which was printed in 1637. In this publication, he also used the letters a , b , and c to represent known quantities. There is a (possibly fictitious) story that, when the book was being printed for the first time, the printer began to run short of the last three letters of the alphabet. So the printer asked Descartes if it mattered which of x , y , or z were used for the mathematical equations in the book. Descartes decided it made no difference to him; so the printer decided to use x predominantly for the mathematics in the book, because the letters y and z would occur more often in the body of the text (written in French) than the letter x would! This might explain why the letter x is still used today as the most common variable to represent unknown quantities in algebra.

Simplifying and Evaluating Algebraic Expressions

To **simplify an expression** means to do all the math possible. For example, to simplify $4 \bullet 2 + 1$ we would first multiply $4 \bullet 2$ to get 8 and then add 1 to get 9. We have introduced most of the symbols and notation used in algebra, but now we need to clarify the order of operations. Otherwise, expressions may have different meanings, and they may result in different values. Consider $2 + 7 \bullet 3$. Do you add first or multiply first? Do you get different answers?

Add first: $9 \bullet 3 = 27$	Multiply first: $2 + 21 = 23$	Which one is correct?
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Early on, mathematicians realized the need to establish some guidelines when performing arithmetic operations to ensure that everyone would get the same answer. Those guidelines are called the order of operations and are listed in the table below.

Step 1: Parentheses and Other Grouping Symbols	Simplify all expressions inside the parentheses or other grouping symbols, working on the innermost parentheses first.
Step 2: Exponents	Simplify all expressions with exponents.
Step 3: Multiplication and Division	Perform all multiplication and division in order from left to right. These operations have equal priority.
Step 4: Addition and Subtraction	Perform all addition and subtraction in order from left to right. These operations have equal priority.

 You may have heard about Please Excuse My Dear Aunt Sally or PEMDAS. Be careful to notice in Steps 3 and 4 in the table above that multiplication and division, as well as addition and subtraction, happen in order from LEFT to RIGHT. It is possible, for example, to have PEDMAS or PEMDSA. The PEMDAS trick can be misleading if not fully understood!

EXAMPLE 5.5

Making a Numerical Equation True Using the Order of Operations

Use parentheses to make the following statements true.

- $17 - 10 + 3 = 10$
- $2 \bullet 26 - 7 = 38$
- $8 + 12 \div 5 - 3 = 14$
- $5 + 2^3 \bullet 7 = 91$

✓ **Solution**

1. Add the parentheses around the $17 - 10$. Then you have $(17 - 10) + 3 = 7 + 3 = 10$.
2. Add the parentheses around the $26 - 7$. Then you have $2 \bullet (26 - 7) = 2 \bullet 19 = 38$.
3. Add the parentheses around the $5 - 3$. Then you have $8 + 12 \div (5 - 3) = 8 + 12 \div 2 = 8 + 6 = 14$.
4. Add the parentheses around the $5 + 2^3$. Then you have $(5 + 2^3) \bullet 7 = (5 + 8) \bullet 7 = 13 \bullet 7 = 91$.

> **YOUR TURN 5.5**

Use parentheses and the order of operations to make each equation true.

1. $24 - 17 - 6 = 13$
2. $3 \cdot 6 + 13 = 31$
3. $12 - 6 \div 5 - 3 = 3$
4. $5 \cdot 3^2 + 5 = 70$

In the last example, we simplified expressions using the order of operations. Now we'll evaluate some expressions—again following the order of operations. To evaluate an expression means to find the value of the expression when the variable is replaced by a given number.

EXAMPLE 5.6

Evaluating and Simplifying an Expression

1. Evaluate $3x + 5$ when $x = 2$.
2. Evaluate $x^2 + 3x + 1$ when $x = 2$.

✓ **Solution**

1. To evaluate, let $x = 2$ in the expression, and then simplify: $3(2) + 5 = 6 + 5 = 11$.
2. To evaluate, let $x = 2$ in the expression, and then simplify: $2^2 + 3(2) + 1 = 4 + 6 + 1 = 11$.

> **YOUR TURN 5.6**

1. Evaluate $5x - 6$ when $x = 3$.
2. Evaluate $x^2 - 6x + 3$ when $x = 3$.

Operations of Algebraic Expressions

Algebraic expressions are made up of **terms**. A term is a constant or the product of a constant and one or more variables. Examples of terms are 7, y , $5x^2$, $9a$, and b^5 . The constant that multiplies the variable is called the **coefficient**. Think of the coefficient as the number in front of the variable. Consider the algebraic expressions $5x^2$, which has a coefficient of 5, and $9a$, which has a coefficient of 9. If there is no number listed in front of the variable, then the coefficient is 1 since $x = 1 \bullet x$.

Some terms share common traits. When two terms are constants or have the same variable and exponent, we say they are **like terms**. If there are like terms in an expression, you can simplify the expression by combining the like terms. We add the coefficients and keep the same variable.

EXAMPLE 5.7

Adding Algebraic Expressions

Add $(x^2 + 4x - 9) + (3x^2 - x + 12)$.

✓ **Solution**

Step 1: Add the terms in any order and get the same result (think: $2 + 3 = 3 + 2$) and drop the parentheses:

$$x^2 + 4x - 9 + 3x^2 - x + 12$$

Step 2: Group like terms together:

$$x^2 + 3x^2 + 4x - x - 9 + 12$$

Step 3: Combine the like terms:

$$4x^2 + 3x + 3$$

YOUR TURN 5.7

1. Add $(2x^2 - 4x + 5) + (3x^2 + x - 12)$.

EXAMPLE 5.8

Subtracting Algebraic Expressions

Subtract $(5x^2 + 4x - 9) - (3x^2 - x + 12)$.

Solution

Step 1: Distribute the negative inside the parentheses (think: $2 - (3 - 4) = 2 - 3 + 4 = -1 + 4 = 3$, which is the correct answer). You cannot just drop the parentheses (for example, $2 - 3 - 4 = -1 - 4 = -5$, which is not correct as we have already verified the answer is 3):

$$5x^2 + 4x - 9 - 3x^2 + x - 12$$

Step 2: Group like terms together:

$$5x^2 - 3x^2 + 4x + x - 9 - 12$$

Step 3: Combine the like terms:

$$2x^2 + x - 21$$

YOUR TURN 5.8

1. Subtract $(x^2 - 4x + 8) - (x^2 + 5x + 12)$.

Before looking at multiplying algebraic expressions we look at the **Distributive Property**, which says that to multiply a sum, first you multiply each term in the sum and then you add the products. For example, $5(4 + 3) = 5(4) + 5(3) = 20 + 15 = 35$ can also be solved as $5(4 + 3) = 5(7) = 35$. If we use a variable, then $5(x + 3) = 5x + 15$.

We can extend this example to $(5 + 2)(4 + 3) = (5)(4) + (5)(3) + (2)(4) + (2)(3) = 20 + 15 + 8 + 6 = 49$, which can also be solved as $(5 + 2)(4 + 3) = (7)(7) = 49$. If we use variables, then $(x + 5)(x + 4) = (x)(x) + (x)(4) + (5)(x) + (5)(4) = x^2 + 4x + 5x + 20 = x^2 + 9x + 20$.

FORMULA

Distributive Property: $a(b + c) = ab + ac$

EXAMPLE 5.9**Simplifying an Expression Using the Order of Operations**

Simplify each expression.

1. $(x - 3)5$
2. $(-3)(x + y - 2)$
3. $5^2(7 + 3)(x)$
4. $4 + x \bullet 5$
5. $(4 + x) \bullet 5$

 Solution

1. $5x - 3 \bullet 5 = 5x - 15$
2. $(-3) \bullet x + (-3) \bullet y - (-3) \bullet 2 = -3x - 3y + 6$
3. $25(7 + 3)(x) = 25(10)(x) = 250x$
4. $4 + 5x$
5. $(4) \bullet (5) + (x) \bullet (5) = 20 + 5x$

 YOUR TURN 5.9

Simplify each expression.

1. $2(y + 5)$
2. $(-2)(a + b - 4)$
3. $4^2(47 - 40 + x)$
4. $(18 \div 3)(x + 7 - 4)$
5. $2(3a + 5) + (-3)(a + 2)$

EXAMPLE 5.10**Multiplying Algebraic Expressions**Multiply $(4x - 9)(x + 2)$. **Solution****Step 1:** Use the Distributive Property:

$$(4x)(x) + (4x)(2) - (9)(x) - (9)(2)$$

Step 2: Multiply:

$$4x^2 + 8x - 9x - 18$$

Step 3: Combine the like terms:

$$4x^2 - x - 18$$

 YOUR TURN 5.10

1. Multiply $(x - 4)(2x - 3)$.

 You may have heard the term FOIL which stands for: First, Outer, Inner, Last. FOIL essentially describes a way to use the Distributive Property if you multiply a two-term expression by another two-term expression, but FOIL only works in that specific situation. For example, suppose you have a two-term expression multiplied by a three-term expression, such as $(x + 2)(x + y - 5)$. What terms qualify as inner terms and what terms qualify as outer terms? In

this particular situation, FOIL cannot possibly work; the multiplication of $(x + 2)(x + y - 5)$ should yield six terms, where FOIL is designed to only give you four! The Distributive Property works regardless of how many terms there are. FOIL can be misleading and applied inappropriately if not fully understood!

EXAMPLE 5.11

Dividing Algebraic Expressions

Divide $(8x^2 + 4x - 16) \div (4x)$.

Solution

Divide EACH term by $4x$:

$$(8x^2 \div 4x) + (4x \div 4x) - (16 \div 4x) = 2x + 1 - \frac{4}{x}$$

YOUR TURN 5.11

1. Divide $(16x^2 + 4x - 8) \div (4)$.



Be careful how you divide! Sometimes students incorrectly divide only one term on top by the bottom term. For example, $\frac{8x^2 + 6x - 3}{2x}$ might turn into $4x + 3x - 3 = 7x - 3$ if done incorrectly. When we divide expressions, EACH term is divided by the divisor. So, $\frac{8x^2 + 6x - 3}{2x} = \frac{8x^2}{2x} + \frac{6x}{2x} - \frac{3}{2x} = 4x + 3 - \frac{3}{2x}$. If you forget, it is always a good idea to check these rules by creating an example using numerical expressions. For example, $\frac{9 + 6 + 3}{3} = \frac{18}{3} = 6$. Dividing each term on top by 3 would yield $\frac{9 + 6 + 3}{3} = \frac{9}{3} + \frac{6}{3} + \frac{3}{3} = 3 + 2 + 1 = 6$, which is the correct answer. However, if you just divided the 9 on top by the 3 on the bottom, getting $\frac{9 + 6 + 3}{3} = 3 + 6 + 3 = 12$, this does not result in the correct answer.



PEOPLE IN MATHEMATICS

Al-Khwarizmi



Figure 5.3 Al-Khwarizmi

Abu Ja'far Muhammad ibn Musa Al-Khwarizmi was born around 780 AD, probably in or around the region of Khwarizm, which is now part of modern-day Uzbekistan. For most of his adult life, he worked as a scholar at the House of Wisdom in Baghdad, Iraq. He wrote many mathematical works during his life, but is probably most famous for his book *Al-kitab al-muhtasar fi hisab al-jabr w'al'muqabalah*, which translates to *The Condensed Book on the Calculation of al-jabr (completion) and al'muqabalah (balancing)*. The word *al-jabr* would eventually become the word we use to describe the topic that he was writing about in this book: *algebra*. From another book of his, with the Latin title *Algoritmi de numero Indorum* (*Al-Khwarizmi on the Hindu Art of Reckoning*), our word *algorithm* is derived. In addition to writing on mathematics, Al-Khwarizmi wrote works on astronomy, geography, the sundial, and the calendar.

In 2012, Andrew Hacker wrote an opinion piece in the *New York Times Magazine* suggesting that teaching algebra in high school was a waste of time. Keith Devlin, a British mathematician, was asked to comment on Hacker's article by his students in his [Stanford University Continuing Studies course "Mathematics: Making the Invisible Visible"](https://openstax.org/r/Making_the_Invisible_Visible) (https://openstax.org/r/Making_the_Invisible_Visible) on iTunes University. Devlin concludes that Hacker was displaying his ignorance of what algebra is.

▶ VIDEO

[Q&A: Why We Teach Algebra \(https://openstax.org/r/Teach_Algebra\)](https://openstax.org/r/Teach_Algebra)

Check Your Understanding

1. Juliette is 2 inches taller than her friend Vivian. Which algebraic equations represent their height? Use J for Juliette's height and V for Vivian's height.

$$J = V + 2$$

$$V = J - 2$$

$$J + 2 = V$$

$$J = V - 2$$

2. Which options represent algebraic expressions?

$$2x^2 + 3x - 1 = 0$$

$$5x + 8$$

$$2n + 3m$$

$$5x - 7 = 3x + 1$$

3. Which expression equals $10x$?

$$(8x + 12x) \div 4x - 2x$$

$$8x + (12x \div 4x) - 2x$$

$$8x + 12x \div (4x - 2x)$$

$$(8x + 12x) \div (4x - 2x)$$

4. Using the expression $3x^2 - 7x + 2$, when a certain number is put in for x , the result is 50. What is the value of x ?

$$-2$$

$$-3$$

$$2$$

$$3$$

5. Which expression equals $(x - y)(x - y)$? Hint: Use the Distributive Property.

$$x^2 - y^2$$

$$x^2 + y^2$$

$$x^2 - 2xy - y^2$$

$$x^2 - 2xy + y^2$$

6. Given the expression $9x^3 + 3x^2 - 6x$, the Distributive Property allows it to be rewritten as:

$$3x(3x^2 + x - 2)$$

$$3x^2 + x - 2$$

$$27x^5 - 54x^4$$

$$27x^6 - 54x^3$$

7. Given the two algebraic expressions $(x + 2)$ and $(x + y - 5)$, the solution is $x^2 + xy - 3x + 2y - 10$. What mathematical operation was performed on the two algebraic expressions?

8. Given the two algebraic expressions $8x^2 - 9x + 6$ and $6x$, the solution is $3x - 1.5 + \frac{1}{x}$. What mathematical operation was performed on the two algebraic expressions?



SECTION 5.1 EXERCISES

For the following exercises, translate from algebra to words.

1. $50 - 15$
2. $(10)(x)$
3. $2a - b$
4. $100 \div 33$
5. $3x + 5$

For the following exercises, translate from words to algebra.

6. 15 divided by 3.
7. The sum of 13 and 13.
8. 120 minus 12.
9. The product of 5 and 4.
10. The sum of double x and 5.

For the following exercises, translate from an English phrase to an expression.

11. Three times y minus 7.
12. a divided by 2; then add 4.
13. x squared minus 3.
14. A rental car company charges \$0.15 per mile m , plus a \$40 flat fee for the rental.
15. A parking garage in New York City charges \$20 for the first hour, then \$5 per hour h .

For the following exercises, use parentheses to make the statements true.

16. $16 \div 4 \cdot 2 + 5 = 13$
17. $2^2 - 5 + 3 \cdot 2 = 5$
18. $x - 3 \cdot x - 2 = x^2 - 5x + 6$

19. $20x \div 5 - 1 - 5x = 0$
 20. $5x + 3x \div 3 - 7x + 1 \cdot x = 0$

For the following exercises, evaluate and simplify the expression.

21. x^2 when $x = 9$
 22. $2x + 5$ when $x = 3$
 23. $(3x + 1)(4x - 6)$ when $x = 2$
 24. $x^2 + 3x + 8$ when $x = 3$
 25. $(x^2 + 5x - 4)(2x)$ when $x = 4$
 26. $4a + 5 - 2a - 8$ when $a = 6$
 27. $8a^2 + 4a + 9 - a^2 - 1$ when $a = 5$
 28. Yasenia is 3 years older than Xavier. How old is Yasenia when Xavier is 18 years old?
 29. A rental car company charges \$0.15 per mile m , plus a \$40 flat fee for the rental. What is the cost of the car rental if one drives 100 miles?
 30. A parking garage in New York City charges \$20 for the first hour, then \$5 per hour h . What is the cost of parking for 10 hours?

For the following exercises, perform the indicated operation for the expressions.

31. Add $(4x - 9) + (x + 12)$.
 32. Add $(3x^2 + 2x + 1) + (x^2 - 2x + 2)$.
 33. Subtract $(4x - 9) - (-x + 2)$.
 34. Subtract $(3x^2 + 5x) - (x^2 - 3x + 11)$.
 35. Multiply $4(x + 2)$.
 36. Multiply $2(3x^2 - 2x + 1)$.
 37. Multiply $(3x)(x - 1)$.
 38. Multiply $(2x - 1)(x + 3)$.
 39. $(125x^2 + 35x - 5) \div (5)$.
 40. $(9x^2 + 18x - 27) \div (3x)$.